Class Activity on 01-08-2026

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Question 1: Footprinting Activity
Target: scanme.nmap.org

Aim:
To perform footprinting on the authorized target scanme.nmap.org using Kali Linux tools and collect publicly available information.

Objective

- Identify the IP address.
- Gather DNS records.
- Discover active hosts.
- Analyze security risks.
- Recommend security measures.

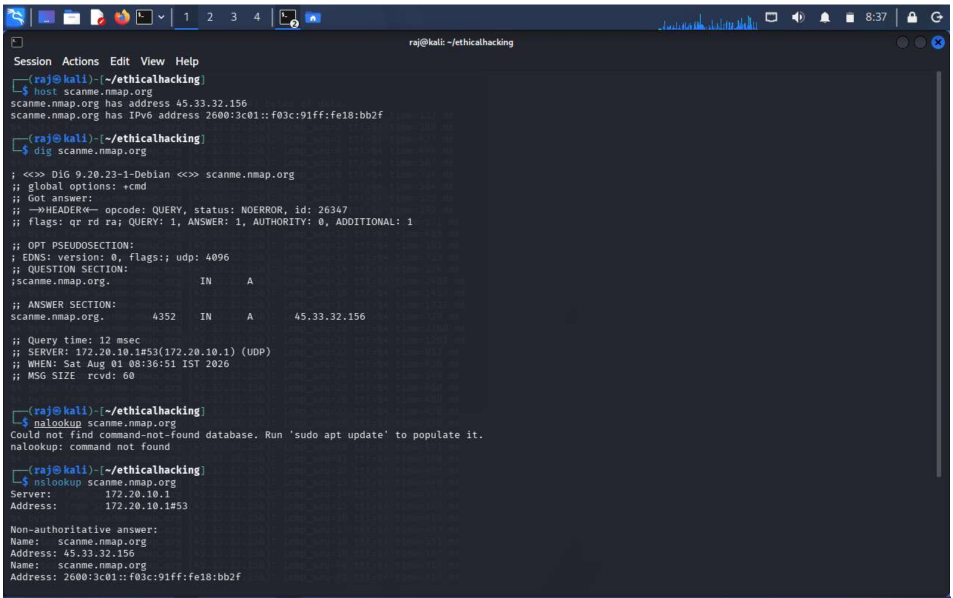
Tools Used

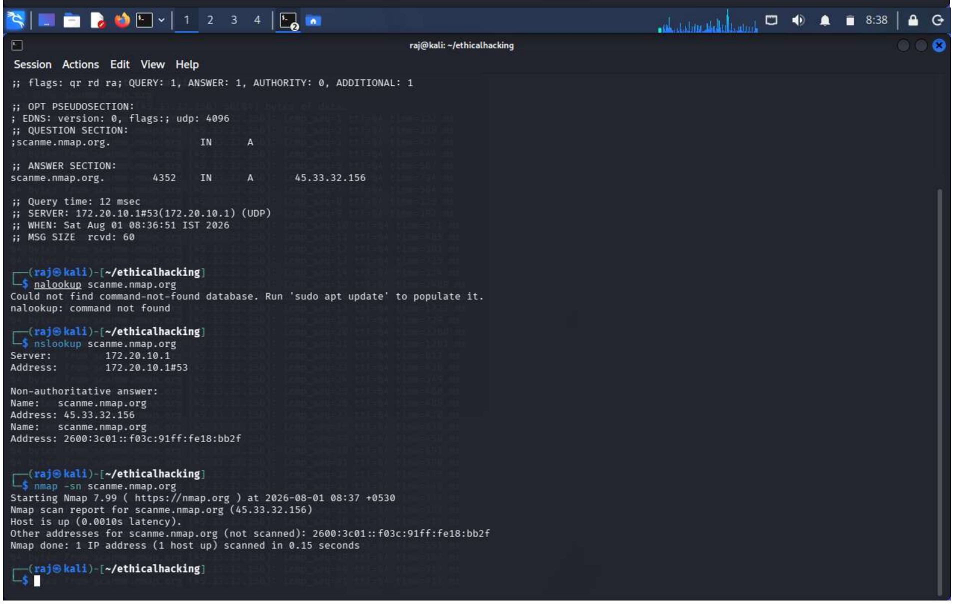
- Kali Linux
- Nmap
- Dig
- Nslookup
- Host
- Ping

Commands Executed

1. Find IP Address
host scanme.nmap.org
2. DNS Information
dig scanme.nmap.org
3. DNS Lookup
nslookup scanme.nmap.org
4. Host Information
host scanme.nmap.org
5. Discover Active Host
nmap -sn scanme.nmap.org

Screenshots of output:





Infrastructure Mapping

Public DNS information may reveal details about how the organization's internet-facing systems are organized, helping an attacker identify areas that warrant further investigation.

Social Engineering Preparation

Knowing legitimate domain names and infrastructure details can help attackers create more convincing phishing emails or fake websites that resemble trusted organizational resources.

Security Recommendations

➢ Limit unnecessary public DNS information.

- Keep DNS records updated.
- Use firewalls to restrict unnecessary services.
- Monitor reconnaissance activities.
- Perform regular security assessments.
- Deploy IDS/IPS solutions.
- Train employees to recognize phishing attempts.

Conclusion

Footprinting provides valuable information about a target using publicly available resources. Organizations should minimize unnecessary information exposure and continuously monitor their external attack surface.

Question 2: Social Engineering Intelligence Gathering
Target: www.wikipedia.org

Aim
To collect publicly available information that demonstrates how attackers could prepare social engineering campaigns.

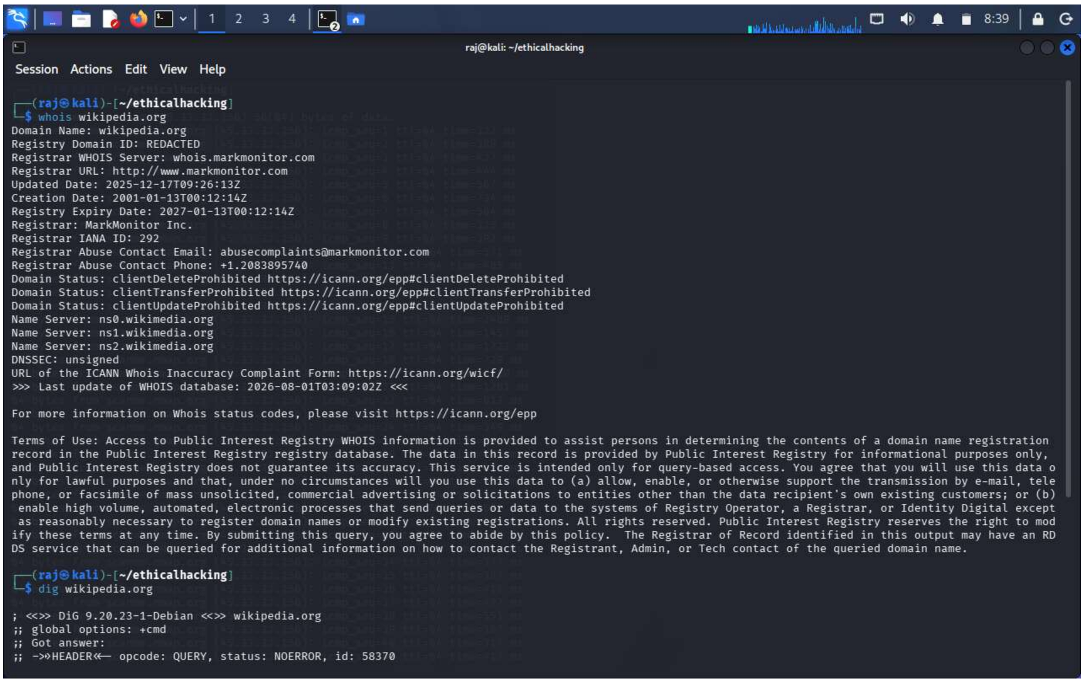
Tools Used

- Web Browser
- WHOIS
- Dig
- Google Search

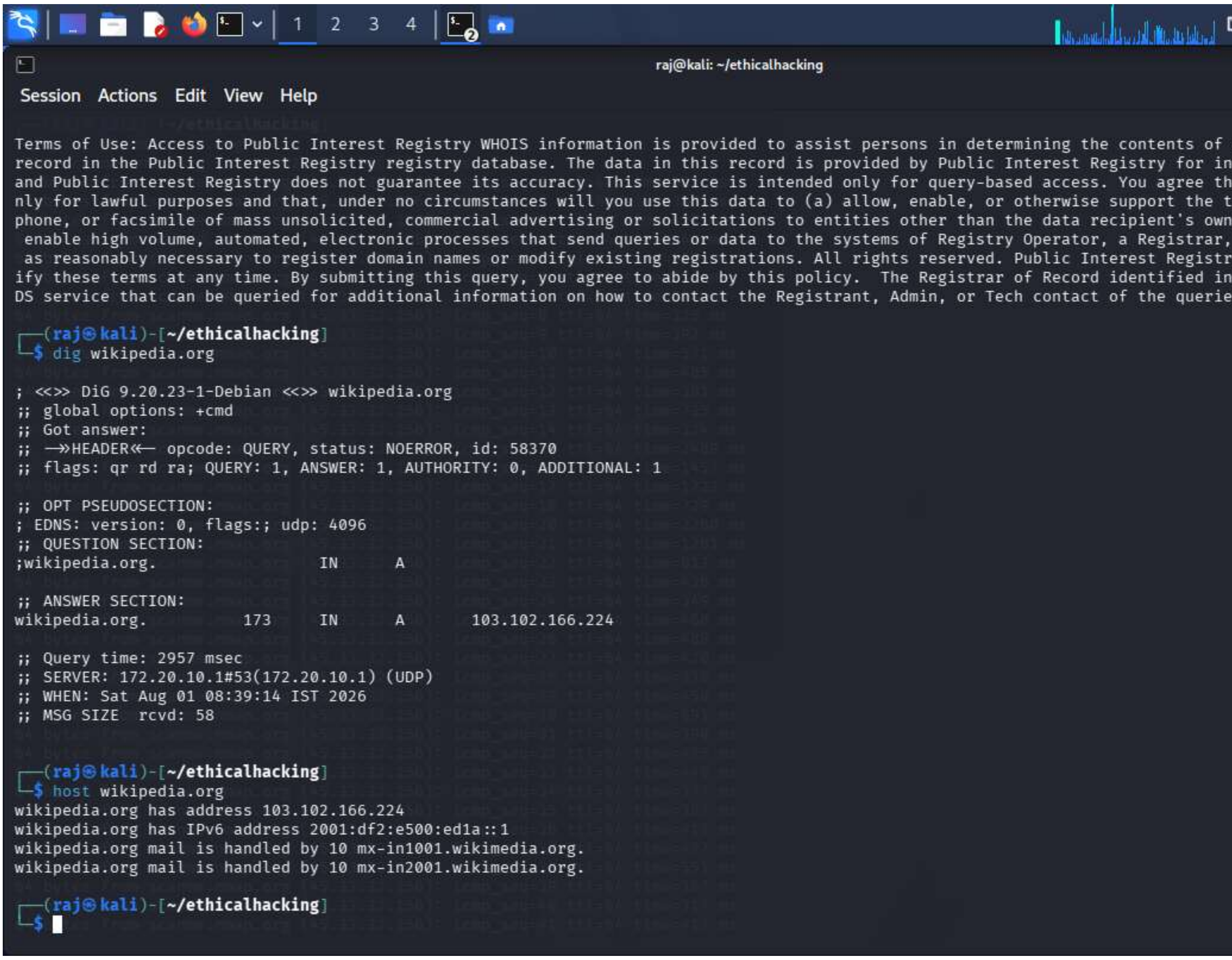
Commands

whois wikipedia.org
dig wikipedia.org
host wikipedia.org

Screenshots of output:



```
raj@kali: ~/ethicalhacking
Session  Actions  Edit  View  Help

┌──(raj㉿kali)-[~/ethicalhacking]
└─$ whois wikipedia.org
Domain Name: wikipedia.org
Registry Domain ID: REDACTED
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2025-12-17T09:26:13Z
Creation Date: 2001-01-13T00:12:14Z
Registry Expiry Date: 2027-01-13T00:12:14Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2083895740
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: ns0.wikimedia.org
Name Server: ns1.wikimedia.org
Name Server: ns2.wikimedia.org
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://icann.org/wicf/
>>> Last update of WHOIS database: 2026-08-01T03:09:02Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

Terms of Use: Access to Public Interest Registry WHOIS information is provided to assist persons in determining the contents of a domain name registration
record in the Public Interest Registry registry database. The data in this record is provided by Public Interest Registry for informational purposes only,
and Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree that you will use this data o
nly for lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the transmission by e-mail, tele
phone, or facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own existing customers; or (b)
 enable high volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar, or Identity Digital except
 as reasonably necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registry reserves the right to mod
ify these terms at any time. By submitting this query, you agree to abide by this policy.  The Registrar of Record identified in this output may have an RD
DS service that can be queried for additional information on how to contact the Registrant, Admin, or Tech contact of the queried domain name.

┌──(raj㉿kali)-[~/ethicalhacking]
└─$ dig wikipedia.org

; <<>> DiG 9.20.23-1-Debian <<>> wikipedia.org
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 58370
```



```
raj@kali: ~/ethicalhacking
Session  Actions  Edit  View  Help

Terms of Use: Access to Public Interest Registry WHOIS information is provided to assist persons in determining the contents of
record in the Public Interest Registry registry database. The data in this record is provided by Public Interest Registry for in
and Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree th
nly for lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the t
phone, or facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own
 enable high volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar,
 as reasonably necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registr
ify these terms at any time. By submitting this query, you agree to abide by this policy.  The Registrar of Record identified in
DS service that can be queried for additional information on how to contact the Registrant, Admin, or Tech contact of the querie

┌──(raj㉿kali)-[~/ethicalhacking]
└─$ dig wikipedia.org

; <<>> DiG 9.20.23-1-Debian <<>> wikipedia.org
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 58370
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;wikipedia.org.                 IN      A

;; ANSWER SECTION:
wikipedia.org.          173     IN      A       103.102.166.224

;; Query time: 2957 msec
;; SERVER: 172.20.10.1#53(172.20.10.1) (UDP)
;; WHEN: Sat Aug 01 08:39:14 IST 2026
;; MSG SIZE  rcvd: 58

┌──(raj㉿kali)-[~/ethicalhacking]
└─$ host wikipedia.org
wikipedia.org has address 103.102.166.224
wikipedia.org has IPv6 address 2001:df2:e500:ed1a::1
wikipedia.org mail is handled by 10 mx-in1001.wikimedia.org.
wikipedia.org mail is handled by 10 mx-in2001.wikimedia.org.

┌──(raj㉿kali)-[~/ethicalhacking]
└─$ 
```

Potential Risks

- Credential theft
- Unauthorized access
- Identity impersonation
- Financial fraud
- Data leakage

Awareness Recommendations

- Conduct phishing awareness training.
- Verify unexpected requests through official channels.
- Encourage reporting of suspicious emails.
- Enable Multi-Factor Authentication (MFA).
- Implement SPF, DKIM, and DMARC.
- Use email security gateways.
- Regularly simulate phishing exercises.

Conclusion

Publicly available information can be combined to make social engineering attempts appear more believable. Security awareness and layered technical controls reduce this risk.

Question 3: Footprinting and Social Engineering Assessment
Target: example.com

Aim

To perform footprinting on example.com and assess how publicly available information could contribute to a hypothetical social engineering scenario.

Tools Used

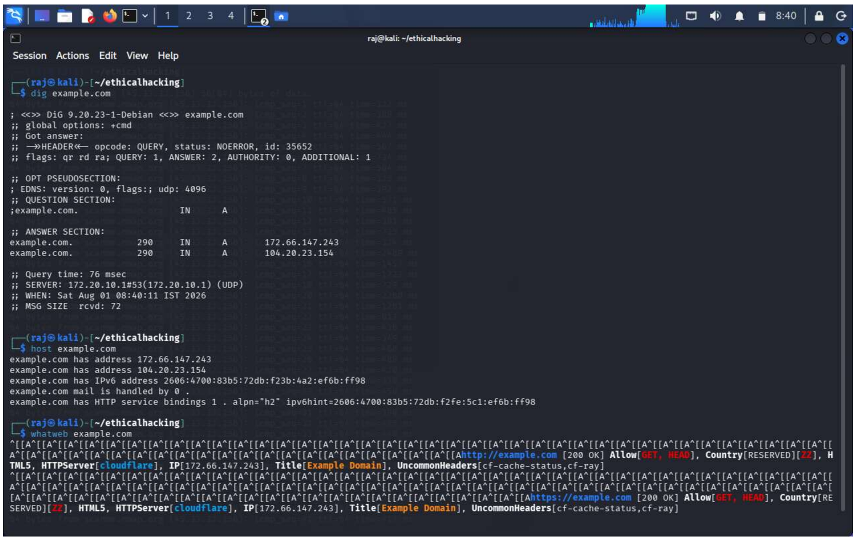
- theHarvester
- Dig
- Host
- Nmap
- WhatWeb

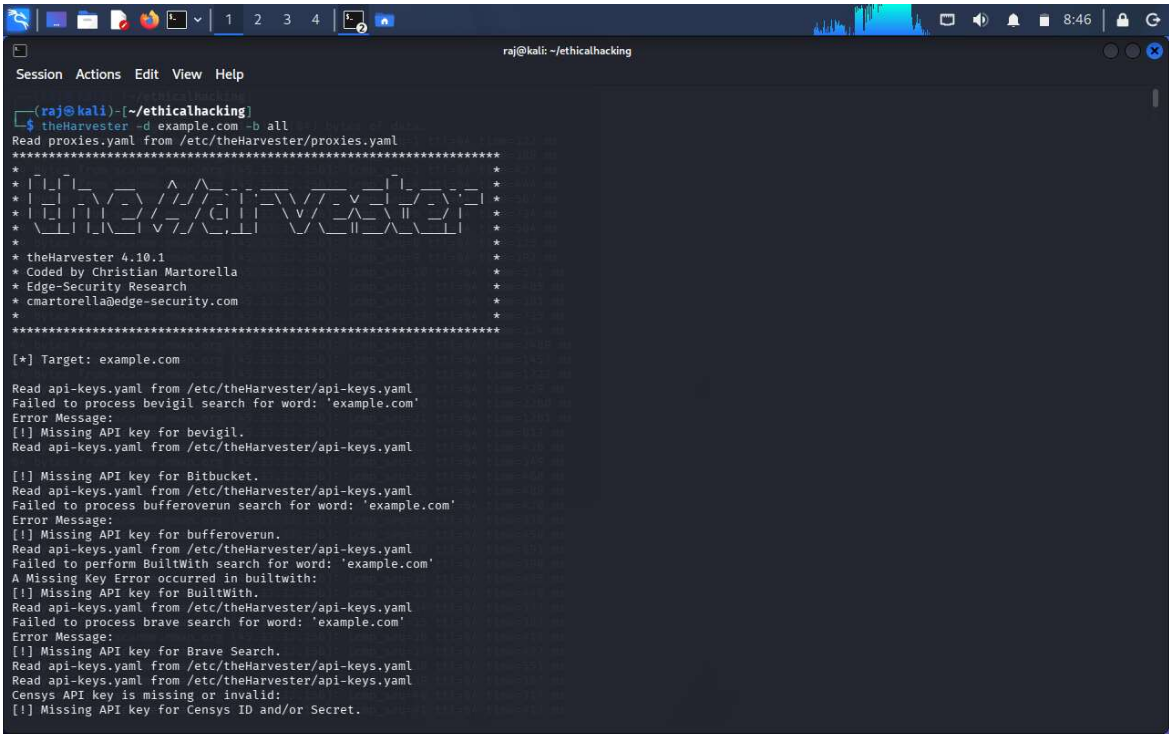
Commands

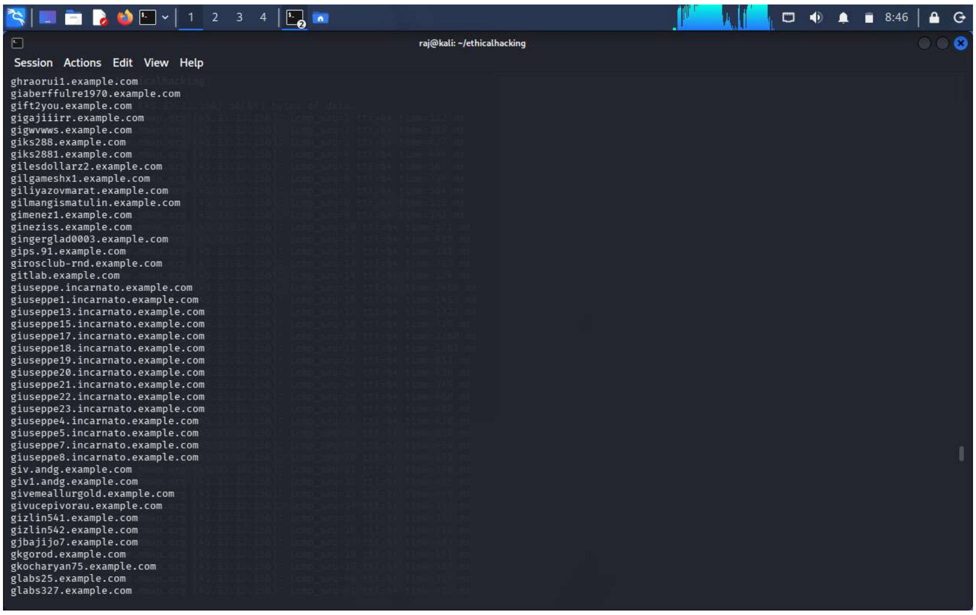
DNS:dig example.com
Host:host example.com
Web Technologies:whatweb example.com

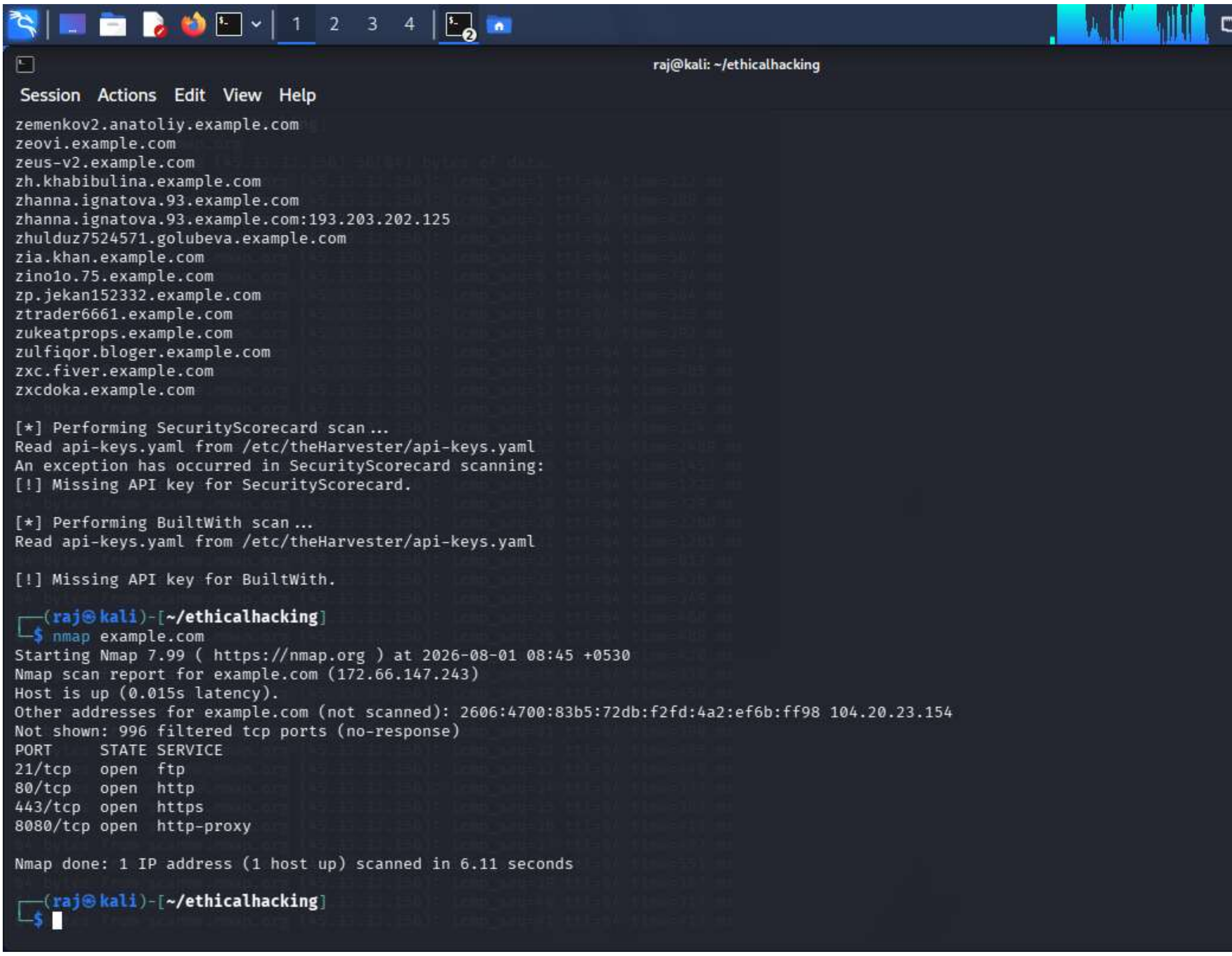
Subdomain Enumeration:theHarvester -d example.com -b all
Port Scan:nmap example.com

Screenshots of output:






raj@kali: ~/ethicalhacking
Session Actions Edit View Help
ghraorui1.example.com
giaberffulre1970.example.com
gift2you.example.com
gigajiiirr.example.com
gigwvwws.example.com
giks288.example.com
giks2881.example.com
gilesdollarz2.example.com
gilgameshx1.example.com
giliyazovmarat.example.com
gilmangismatulin.example.com
gimenez1.example.com
gineziss.example.com
gingerglad0003.example.com
gips.91.example.com
girosclub-rnd.example.com
gitlab.example.com
giuseppe.incarnato.example.com
giuseppe1.incarnato.example.com
giuseppe13.incarnato.example.com
giuseppe15.incarnato.example.com
giuseppe17.incarnato.example.com
giuseppe18.incarnato.example.com
giuseppe19.incarnato.example.com
giuseppe20.incarnato.example.com
giuseppe21.incarnato.example.com
giuseppe22.incarnato.example.com
giuseppe23.incarnato.example.com
giuseppe4.incarnato.example.com
giuseppe5.incarnato.example.com
giuseppe7.incarnato.example.com
giuseppe8.incarnato.example.com
giv.andg.example.com
giv1.andg.example.com
givemeallurgold.example.com
givucepivorau.example.com
gizlin541.example.com
gizlin542.example.com
gjbajijo7.example.com
gkgorod.example.com
gkocharyan75.example.com
glabs25.example.com
glabs327.example.com


raj@kali: ~/ethicalhacking
Session Actions Edit View Help
zemenkov2.anatoliy.example.com
zeovi.example.com
zeus-v2.example.com
zh.khabibulina.example.com
zhanna.ignatova.93.example.com
zhanna.ignatova.93.example.com:193.203.202.125
zhulduz7524571.golubeva.example.com
zia.khan.example.com
zino1o.75.example.com
zp.jekan152332.example.com
ztrader6661.example.com
zukeatprops.example.com
zulfiqor.bloger.example.com
zxc.fiver.example.com
zxcdoka.example.com
[*] Performing SecurityScorecard scan ...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.
[*] Performing BuiltWith scan ...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.
(raj㉿kali)-[~/ethicalhacking]
$ nmap example.com
Starting Nmap 7.99 (https://nmap.org) at 2026-08-01 08:45 +0530
Nmap scan report for example.com (172.66.147.243)
Host is up (0.015s latency).
Other addresses for example.com (not scanned): 2606:4700:83b5:72db:f2fd:4a2:ef6b:ff98 104.20.23.154
Not shown: 996 filtered tcp ports (no-response)
PORT STATE SERVICE
21/tcp open ftp
80/tcp open http
443/tcp open https
8080/tcp open http-proxy
Nmap done: 1 IP address (1 host up) scanned in 6.11 seconds
(raj㉿kali)-[~/ethicalhacking]
$

Potential Impact

- Compromise of user credentials
- Unauthorized access to internal systems
- Exposure of sensitive data
- Financial loss
- Reputational damage

Countermeasures

- Implement Multi-Factor Authentication (MFA).
- Conduct regular phishing awareness training.
- Deploy email authentication (SPF, DKIM, DMARC).
- Limit unnecessary public information exposure.
- Monitor for lookalike domains and phishing websites.
- Apply the principle of least privilege.
- Keep systems and web applications updated.
- Use web application firewalls and endpoint protection.
- Encourage prompt reporting of suspicious communications.
- Perform regular external security assessments.

Conclusion

Footprinting helps defenders understand what information about their organization is publicly visible. Reducing unnecessary exposure, maintaining strong authentication, and training users to recognize phishing attempts are key measures to mitigate social engineering risks.